THE H-COHOMOLOGY OF THE PERFECTED STRUCTURE SHEAF IS THE COLIMIT OF THE COHOMOLOGIES OF THE STRUCTURE SHEAF

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file etale-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the proof is written out in full: the identification through the flatness-chapter lemma, the essential-image argument, the Leray chain, and the colimit computation of the fppf cohomology.

Editorial note. The label proposition-h-cohomology-structure-sheaf names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own proposition environment in the chapter source.

Editorial note. The statement is an author Proposition environment, delivered verbatim, including the author's own remark on the notation; the five cited inputs are kept as external pointers with their canonical Stacks tags.

Allowed standard prerequisites (editorial).

- The perfection of the structure sheaf in characteristic p. More on Flatness, the lemma identifying the h-sheafification of the structure sheaf with the perfection colim_F O. Supplies the identity O^h = O^perf = colim_F O.
- Theperfection lies in the essential image of the h-pushforward. Same chapter, the lemma placing O^perf in the essential image of R epsilon_* from the h-topology. Supplies the object K with R epsilon_*K = O^perf and hence the identification R epsilon_*O^perf = O^perf.
- Leray for the comparison morphism and cohomology of a filtered colimit. Cohomology on Sites, the comparison section for the Leray spectral sequence, together with the computation of fppf cohomology of a filtered colimit. Supplies the chain of identities computing the h-cohomology.

Source reference. Original lemma 0EVW: sources/stacks-chow/tags.txt, source lines 18601–18601. Prerequisite pointer invoked at proof line(s) 23723; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0EVK: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 23651–23651. Prerequisite pointer invoked at proof line(s) 23720; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

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Source reference. Original section 0EWK: sources/stacks-chow/tags.txt, source lines 18625–18625. Prerequisite pointer invoked at proof line(s) 23729; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Proposition 0.1. *Let p be a prime number. Let S be a quasi-compact and quasi-separated scheme over $\mathbf{F}_p$. Then*

$$H^i((Sch/S)_h, \mathcal{O}^h) = \operatorname{colim}_F H^i(S, \mathcal{O})$$

Here on the left hand side by $\mathcal{O}^h$ we mean the h sheafification of the structure sheaf.

Proof. This is just a reformulation of Lemma 0EVK. Recall that $\mathcal{O}^h = \mathcal{O}^{perf} = \operatorname{colim}_F \mathcal{O}$, see More on Flatness, Lemma 0EVW. By Lemma 0EVK we see that $\mathcal{O}^{perf}$ viewed as an object of $D((Sch/S)_{fppf})$ is of the form $R\epsilon_* K$ for some $K \in D((Sch/S)_h)$. Then $K = \epsilon^{-1}\mathcal{O}^{perf}$ which is actually equal to $\mathcal{O}^{perf}$ because $\mathcal{O}^{perf}$ is an h sheaf. See Cohomology on Sites, Section 0EWK. Hence $R\epsilon_*\mathcal{O}^{perf} = \mathcal{O}^{perf}$ (with apologies for the confusing notation). Thus the lemma now follows from Leray

$$R\Gamma_h(S, \mathcal{O}^{perf}) = R\Gamma_{fppf}(S, R\epsilon_*\mathcal{O}^{perf}) = R\Gamma_{fppf}(S, \mathcal{O}^{perf})$$

and the fact that

$$H^i_{fppf}(S, \mathcal{O}^{perf}) = H^i_{fppf}(S, \operatorname{colim}_F \mathcal{O}) = \operatorname{colim}_F H^i_{fppf}(S, \mathcal{O})$$

as S is quasi-compact and quasi-separated (see proof of Lemma 0EVK). □